NALLABATI VENKATA DURGA SAI SARAT CHANDRA

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EDUCATION & ACADEMIC PEDIGREE

Amrita Vishwa Vidyapeetham — Amritapuri, India
Bachelor of Technology in Robotics and Artificial Intelligence — **2022 – 2026 (Expected)**
• **Cumulative Grade Point Average: 9.05 / 10.00** • **Class Rank:** Top 2%
• **Core Coursework:** Multivariable Calculus, Linear Algebra, Real Analysis, Partial Differential Equations, Non-Newtonian Fluid Mechanics, Classical Mechanics, Kinematics & Dynamics, Optimal Control, Digital Signal Processing, Embedded Systems, Deep Learning, Autonomous Navigation.

State Board of Technical Education and Training (SBTET) — Andhra Pradesh, India
Diploma in Mechanical Engineering — **2019 – 2022**
• **Performance: 92.4%** • **State Rank: AIR 497** (Top 0.4% among 125,000+ candidates in AP POLYCET).

PEER-REVIEWED PUBLICATIONS & SCIENTIFIC MANUSCRIPTS

[J1] Empirical Performance Analysis and Economic Assessment of a Concentrated Solar Still Elsevier Format (In Review)
N.V.D.S. Sarat Chandra, Lead Researcher
• Executed a 25-day empirical coastal desalination study utilizing a focal parabolic thermal receiver reaching 116°C core temperature.
• Achieved 99.5% Total Dissolved Solids (TDS) rejection (reducing raw seawater from 3,850 to 18 mg/L) with Adjusted $R^2 = 0.83$ ($p < 0.05$).
• Quantified Levelized Cost of Water (LCOW) at $0.034/L, establishing an off-grid thermodynamic solar purification benchmark.

[C1] Real-Time Vision Pipeline for Autonomous Mobile Robots with Multi-Threaded Telemetry — **IEEE ICRM 2025**
N.V.D.S. Sarat Chandra, Autonomous Systems Lead
• Formulated an OpenCV Probabilistic Hough Line Transform pipeline executing at 35+ FPS on ARM Cortex-A CPU compute budgets.
• Integrated asynchronous UDP/IP multi-threading to mitigate camera latency bottlenecks ($<$ 15 ms telemetry overhead).

RESEARCH EXPERIENCE & COMPUTATIONAL SYSTEMS

Scientific Machine Learning: Hemodynamic Flow Reconstruction via PINNs — **2024 – 2025**
• Inverted non-Newtonian blood rheology (Carreau-Yasuda and Casson constitutive models) from pulsatile velocity fields using PyTorch.
• Implemented adaptive Neural Tangent Kernel (NTK) loss balancing, achieving $<$ 2.0% relative L_2 error against ANSYS Fluent CFD ground truth.

Reduced-Order Modeling (ROM) for Navier-Stokes Equations via POD-Galerkin — **2024**
• Projected high-dimensional fluid velocity snapshots onto 12 dominant Proper Orthogonal Decomposition (POD / SVD) spatial modes.
• Preserved 99.2% kinetic energy while accelerating transient numerical simulation runtimes by $1,200\times$.

Evolution Edge: Quantized Neural Inference Architecture — **2024 – 2025**
• Deployed INT8 quantized neural models to AMD Ryzen AI NPUs via ONNX Runtime (Vitis AI EP) with sub-28.4ms latency within an 8W TDP.
• Engineered confidence-based dynamic escalation bridges routing uncertain anomaly patterns to high-compute cloud GPU clusters.

ROBOTICS & EMBEDDED SYSTEMS ARCHITECTURE

CLUB MIRAI: Distributed Companion Robot Control Architecture — **2024 – Present**
• Engineered a 50 Hz ROS 2 Jazzy distributed node network utilizing CycloneDDS zero-copy shared memory communication.
• Formulated a hybrid analytical and Damped Least Squares (DLS) Jacobian Inverse Kinematics solver for 5-DOF trajectory tracking.

Industrial Off-Grid Telemetry Node with 4G LTE/GSM Failover — **2023 – 2024**
• Designed a non-blocking AT command finite state machine with ring buffers on Valetron A7670C hardware for off-grid automation.
• Integrated opto-isolated 240V relay switching with hardware watchdogs, achieving $>$ 99% uptime in agricultural environments.

COMPREHENSIVE TECHNICAL TAXONOMY

- **Scientific ML & AI:** PyTorch, PINNs, Proper Orthogonal Decomposition (POD), SINDy, Reduced-Order Modeling, Optuna, ONNX.
- **Robotics & Middleware:** ROS 2 (Jazzy/Humble), CycloneDDS, Gazebo Sim, MoveIt 2, Inverse Kinematics, OpenCV, Sensor Fusion (EKF).
- **Firmware & Hardware:** Bare-Metal C (C99/C11), Modern C++ (17/20), FreeRTOS, STM32 (ARM Cortex-M), ESP32, DMA Circular UART, CAN 2.0B, I2C, SPI.
- **Engineering Tools:** Linux/Ubuntu Server, Git/GitHub Actions, ANSYS Fluent CFD, SolidWorks, KiCAD, Tectonic LaTeX.

HONORS, AWARDS & NATIONAL FELLOWSHIPS

- **ISRO YUVIKA Space Science Program (2019):** Selected among top 0.1% candidates nationwide for space science immersion by ISRO.
- **National Anveshika Experimental Physics Test (NAEST) – Prof. H.C. Verma:** Qualified 2 competitive national screening stages.
- **IEEE IES GenAI Autonomous Challenge Finalist (2024):** Ranked top finalist for physics-guided diffusion modeling.
- **National Mathematics Olympiad (MTSO):** State Rank 3 in advanced mathematical reasoning.